

Foundation (Jalapeno)

Q1. Factor the trinomial $x^2 + 5x + 6$

- (A) $(x + 2)(x + 3)$
- (B) $x(x + 5) + 6$
- (C) $(x + 1)(x + 4)$
- (D) $(x + 3)(x + 2)$
- (E) $x(x + 6) + 5$

Q2. Factor the trinomial $x^2 - 7x + 12$

- (A) $x(x - 7) + 12$
- (B) $x(x - 3) - 4$
- (C) $(x - 3)(x - 4)$
- (D) $(x - 4)(x - 3)$
- (E) $x(x + 12) - 7$

Q3. A basketball player scored $x^2 + 4x + 4$ points. Factor the expression

- (A) $x(x + 4) + 4$
- (B) $(x + 2)^2$
- (C) $(x + 2)(x - 2)$
- (D) $(x - 2)^2$
- (E) $x(x - 4) + 4$

Q4. Factor the trinomial $2x^2 + 5x - 3$

- (A) $(2x + 3)(x - 1)$
- (B) $(x + 3)(2x - 1)$
- (C) $(2x + 1)(x - 3)$
- (D) $(2x - 1)(x + 3)$
- (E) $(2x - 3)(x + 1)$

Q5. A cyclist rides $x^2 - 2x - 3$ miles. Factor the expression

- (A) $(x + 1)(x - 3)$
- (B) $x(x - 2) - 3$
- (C) $(x - 3)(x + 1)$
- (D) $(x + 3)(x - 1)$
- (E) $(x - 1)(x + 3)$

Q6. Factor the trinomial $3x^2 + 2x - 1$

- (A) $(3x - 1)(x + 1)$
- (B) $(3x + 1)(x - 1)$
- (C) $(x + 1)(3x - 1)$
- (D) $(x - 1)(3x + 1)$
- (E) $(3x + 2)(x - 1)$

Q7. A football team scored $x^2 + x - 6$ points. Factor the expression

- (A) $(x + 3)(x - 2)$
- (B) $(x + 2)(x - 3)$
- (C) $(x - 3)(x + 2)$
- (D) $(x - 2)(x + 3)$
- (E) $x(x + 1) - 6$

Q8. Factor the trinomial $2x^2 + x - 1$

- (A) $(2x - 1)(x + 1)$
- (B) $(2x + 1)(x - 1)$
- (C) $(x + 1)(2x - 1)$
- (D) $(x - 1)(2x + 1)$
- (E) $(x + 2)(2x - 1)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $4x^2 + 12x + 8$. Show all steps and justify your answer.

Q10. A tennis player serves the ball at a speed of $x^2 + 2x - 3$ miles per hour. Factor the expression and explain the factors in the context of the problem.

Chef's Tips

- Check for correct factoring
- Ensure students justify their answers
- Provide feedback on mathematical reasoning